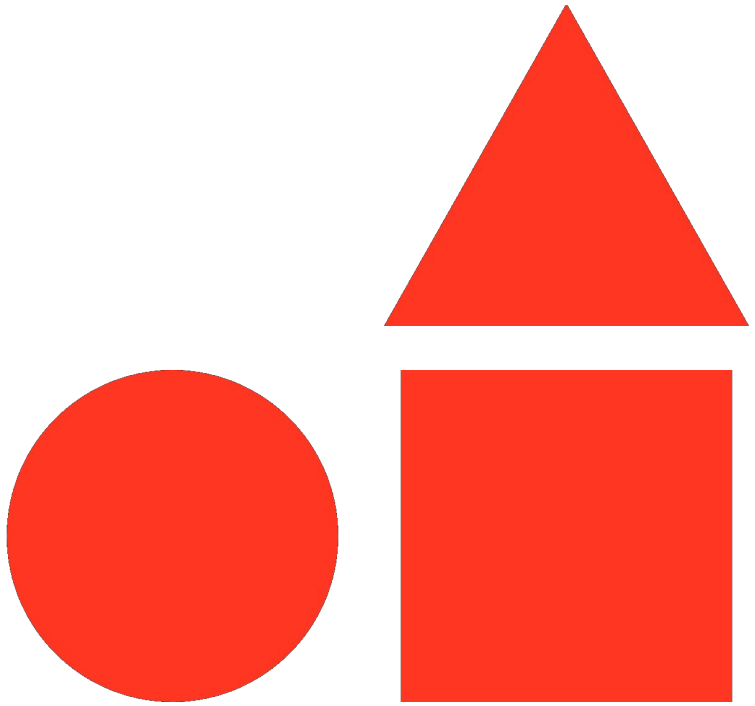
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# Virgin Atlantic Cargo Yield Agent

Turning every RFQ into a capacity-aware, margin-optimal quote in seconds



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Cargo revenue is made or lost one RFQ at a time

# The problem

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# Manual RFQ pricing: slow, inconsistent, capacity-blind

- Forwarders fire the same lane RFQ to every airline — the first good rate back wins the booking
- A human analyst hand-assembles load factors, rate history, competitor intel and IATA rules for every quote
- AOG and pharma need a confirmed rate in under 30 minutes — impossible on a Monday inbox of 80 RFQs
- Manual quoting drifts: two analysts price the same LHR–JFK pharma booking differently → margin leakage

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# What the status quo costs

| KPI | Status quo | Why it hurts |
|---|---|---|
| RFQ turnaround | 15–40 min, manual | AOG/pharma bookings lost to faster carriers |
| Yield consistency | ±10–15% analyst variance | Margin leakage on every quote |
| Capacity utilisation | priced “by gut” | belly flown empty or premium given away |
| Analyst capacity | ~80 RFQs/day ceiling | can’t scale with demand peaks |

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# The solution — a 5-step Cargo Yield Agent

- Capacity check: how full is the target flight; how does this booking move the load factor?
- Yield calc: an ML model predicts the base rate (lane × commodity × forwarder × season), bounded by 90-day actuals
- Competitive check: compares the rate to scraped competitor pricing on the same lane/commodity
- Rules retrieval: pulls IATA / airline handling rules from a vector-indexed knowledge base
- Quote drafting: an LLM writes the quote with a rationale the analyst signs off or overrides
- Every run emits an MLflow trace: why £3.23/kg and not £3.10



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# One integrated Databricks journey

- Lakeflow: ingests raw AWBs, flights, forwarders and competitor rates with data-quality expectations
- Unity Catalog: governs bronze → silver → gold plus the ML and agent artifacts
- Lakebase (Postgres): serves the live RFQ inbox and captures submitted quotes (OLTP)
- ML + GenAI: a yield model and a 5-step agent make it intelligent
- Genie: makes the cargo book queryable in plain English
- Databricks App: the analyst's cockpit. One platform, one governance model, not six stitched-together tools

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# Business outcomes

| Outcome | Before | With the agent |
|---|---|---|
| RFQ response time | 15–40 min | ~1.4 sec |
| Quotes cleared | ~80/day ceiling | agent clears the inbox; humans handle exceptions |
| Yield consistency | ±10–15% | deterministic, auditable, every time |
| Capacity discipline | by gut | auto-premium on tight flights, discount soft ones |
| Incremental margin | — | £2.5–5M / yr from a 1–2% yield lift |

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# Value to each buyer

- VP Cargo Revenue (executive sponsor): protects and grows margin on the existing £236M book; wins time-critical AOG/pharma bookings; the business case is a hard number, not “efficiency”
- Head of Yield / Cargo Data Lead (domain owner): every quote is explainable and overridable; the model and rules are versioned and governed in Unity Catalog; no new data silo to maintain

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# Decisions & trade-offs

- A tool-using agent, not a black-box price model — analysts must see and override the reasoning (MLflow trace)
- Lakebase for OLTP (RFQ inbox + quotes), lakehouse for analytics — each engine used for what it's best at
- ML base rate bounded by 90-day actuals — stops drift to unrealistic rates on thin lanes
- Genie for exploration, the agent for the transaction — right tool per job
- Analytics anchored to the latest data date — KPI tiles always populate whenever the demo is shown

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